Gain selection with a third-order actuator

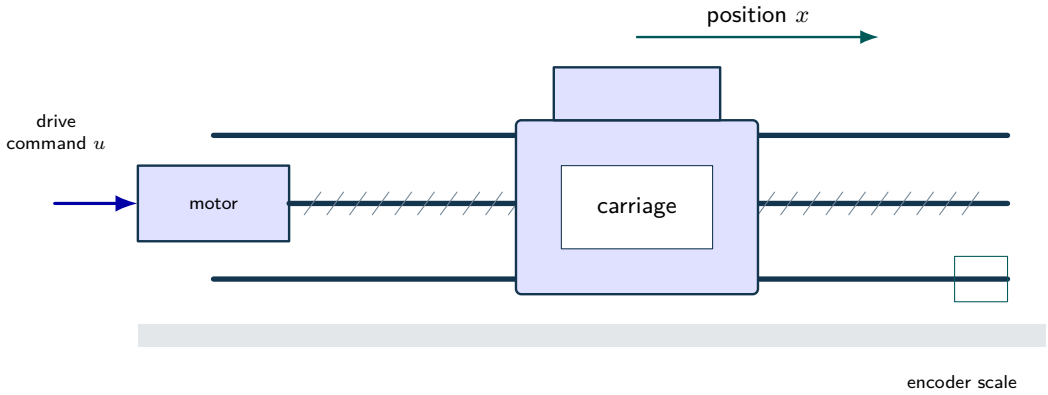
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

A normalized actuator has $G(s) = 1/[s(s+2)(s+6)]$ under unity negative feedback with proportional gain $K > 0$. Derive the strictly stable gain interval using Routh–Hurwitz, the imaginary-axis crossing and real-axis breakaway. Simulate $K = 12$, 60 and 100 to compare regulation and instability. A short time plot alone is not a stability test.

2 Physical system



Illustrative positioning axis; $G(s) = 1/[s(s+2)(s+6)]$ is stipulated

A motorized positioning axis illustrates the role of the integrator and two stable lags. The normalized transfer function is specified as a teaching model; the pictured dimensions do not identify its coefficients.

3 Characteristic equation

The unity-feedback denominator is

$$p(s) = s(s+2)(s+6) + K = s^3 + 8s^2 + 12s + K.$$

The first column of the Routh array is $1, 8, (96-K)/8, K$. Strict stability therefore requires $0 < K < 96$. At $K = 0$ there is a pole at the origin; $K = 96$ is a boundary, not a stable maximum.

4 Imaginary-axis crossing

At $K = 96$, $p(s) = (s+8)(s^2+12)$, so the crossing pair is $s = \pm j\sqrt{12}$. The remaining pole stays at -8.

5 Root-locus geometry

There are three open-loop poles and no finite zeros. The centroid is $-8/3$ and the asymptote angles are 60, 180 and 300 degrees. Real-axis locus segments are $(-\infty, -6)$ and $(-2, 0)$. Since $K(s) = -s(s+2)(s+6)$, a stationary point satisfies $3s^2 + 16s + 12 = 0$. The root $(-8+2\sqrt{7})/3 \simeq -0.9028$ belongs to $(-2, 0)$ and is the valid positive-gain breakaway. The other stationary point does not lie on a positive-gain real-axis segment.

6 Checking the plots

Use pole real parts to classify each gain before plotting. A finite horizon can conceal instability when an unstable real part is small. For stable positive gains, the integrator in the open loop gives zero constant-reference error, but says nothing about overshoot.

7 What to learn from the experiment

An algebraic stability interval provides a guardrail for numerical tuning. The endpoint K=96 must be excluded from strict asymptotic stability.

8 System sketch

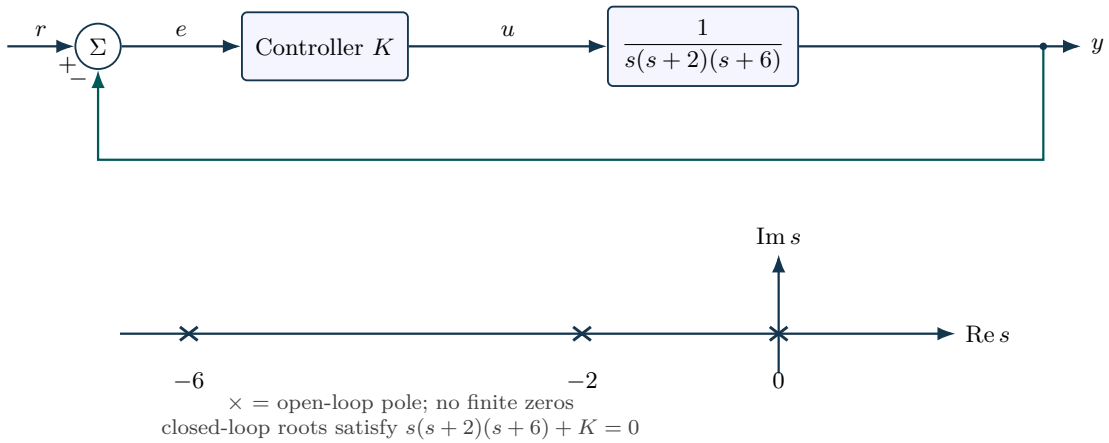
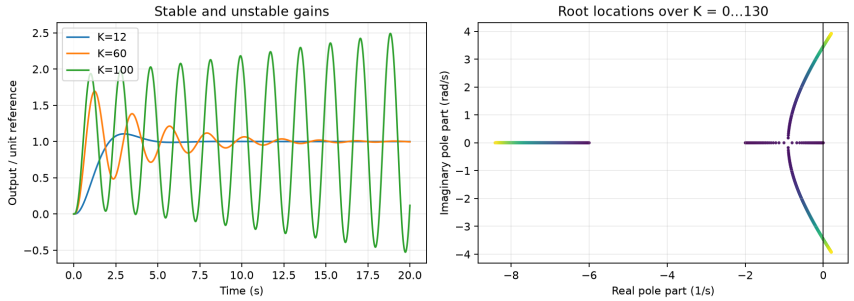


Figure 1. Unity negative feedback and the open-loop pole map. Pole symbols provide the starting geometry for the root-locus exercises.

9 Executed computational evidence

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K=12: largest pole real part=-0.788757
K=60: largest pole real part=-0.265547
K=100: largest pole real part=0.026030
Breakaway=-0.90283246; gain=5.04904248; boundary frequency=3.46410162 rad/s
```



Method references. The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.